Lab Exercise 17- Creating Secrets in Kubernetes

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Batch-2

Creating secrets in Kubernetes is an essential practice for managing sensitive information such as passwords, OAuth tokens, and SSH keys. Let's go through a lab exercise that demonstrates how to create different types of secrets in Kubernetes.

Step 1: Create a Generic Secret

Create a secret from literal values:

```
kubectl create secret generic my-secret --from-literal=username=admin --from-literal=password='P@ssw0rd'
```

```
PS D:\kubs-misha> kubectl create secret generic my-secret --from-literal=username=admin --from-literal=password='P@ssw0r
d'
secret/my-secret created
```

Verify the secret:

```
kubectl get secrets my-secret -o yaml
```

```
PS D:\kubs-misha> kubectl get secrets my-secret -o yaml
apiVersion: v1
data:
  password: UEBzc3cwcmQ=
  username: YWRtaW4=
kind: Secret
metadata:
  creationTimestamp: "2026-03-17T09:05:20Z"
  name: my-secret
  namespace: default
  resourceVersion: "7195"
  uid: 4040e4c7-a352-4931-b5db-9cf6238790e1
type: Opaque
```

The output will be base64 encoded. To decode, you can use:

echo -n “P@ssw0rd” | base64

```
PS D:\kubs-misha> [Convert]::ToBase64String([Text.Encoding]::UTF8.GetBytes("P@ssw0rd"))
UEBzc3cwcmQ=
PS D:\kubs-misha> [System.IO.File]::WriteAllText("username.txt", "my-username")
```

Step 2: Create a Secret from a File

Create a file with your sensitive data:

echo -n 'my-username' > ./username.txt

echo -n 'my-password' > ./password.txt

```
PS D:\kubs-misha> [System.IO.File]::WriteAllText("D:\kubs-misha\username.txt", "my-username")
PS D:\kubs-misha> [System.IO.File]::WriteAllText("D:\kubs-misha\password.txt", "P@ssw0rd")
PS D:\kubs-misha> ls
```

Create a secret from the file:

kubectl create secret generic my-file-secret --from-file=username=./username.txt --from-file=password=./password.txt

```
PS D:\kubs-misha> kubectl create secret generic my-file-secret --from-file=username=./username.txt --from-file=password=
./password.txt
secret/my-file-secret created
```

Verify the secret:

kubectl get secrets my-file-secret -o yaml

```
PS D:\kubs-misha> kubectl get secrets my-file-secret -o yaml
apiVersion: v1
data:
  password: UEBzc3cwcmQ=
  username: bXktdXNlcm5hbWU=
kind: Secret
metadata:
  creationTimestamp: "2026-03-17T09:12:23Z"
  name: my-file-secret
  namespace: default
  resourceVersion: "7778"
  uid: aab52b91-ce0d-4a8f-bfe3-e33f6678a5bd
type: Opaque
```

Step 3: Create a Docker Registry Secret

Create a Docker registry secret:

kubectl create secret docker-registry my-registry-secret --docker-username=<your-username> --docker-password=<your-password> --docker-email=<your-email> --docker-server=<your-docker-server>

```
PS D:\kubs-misha> kubectl create secret docker-registry my-registry-secret --docker-username=<your-username> --docker-pa
ssword=<your-password> --docker-email=<your-email> --docker-server=<your-docker-server>
secret/my-registry-secret created
```

Verify the secret:

kubectl get secrets my-registry-secret -o yaml

```
PS D:\kubs-misha> kubectl get secrets my-registry-secret -o yaml
apiVersion: v1
data:
  .dockerconfigjson: eyJhdXRocyI6eyJcdTAwM2N5b3VyLWRvY2tlci1zZXJ2ZXJcdTAwM2UiOnsidXNlcm5hbWUiOiJcdTAwM2N5b3VyLXVzZX
lXHUwMDNlIiwicGFzc3dvcmQiOiJcdTAwM2N5b3VyLXBhc3N3b3JkXHUwMDNlIiwiZW1haWwiOiJcdTAwM2N5b3VyLWVtYWlsXHUwMDNlIiwiYXV0aC
IbHZkWEl0ZFhObGNtNWhiV1UrT2p4NWIzVnlMWEJoYzNOM2IzSmtQZz09In19fQ==
kind: Secret
metadata:
  creationTimestamp: "2026-03-17T09:12:41Z"
  name: my-registry-secret
  namespace: default
  resourceVersion: "7809"
  uid: 0eeb26d3-32cb-4780-b9a0-61a9ab8cc867
type: kubernetes.io/dockerconfigjson
```

Step 4: Create a Secret Using a YAML Manifest

Create a YAML file for the secret:

Create a file named **my-secret.yaml** with the following content:

```
apiVersion: v1
kind: Secret
metadata:
  name: my-manifest-secret
type: Opaque
data:
  username: YWRtaW4=    # base64 encoded value of 'admin'
  password: UEBzc3cwcmQ=  # base64 encoded value of 'P@ssw0rd'
```

Apply the YAML file:

```
kubectl apply -f my-manifest-secret.yaml
```

```
PS D:\kubs-misha> kubectl apply -f my-secret.yaml
secret/my-manifest-secret created
```

Verify the secret:

```
kubectl get secrets my-manifest-secret -o yaml
```

```
PS D:\kubs-misha> kubectl get secrets my-secret -o yaml
apiVersion: v1
data:
  password: UEBzc3cwcmQ=
  username: YWRtaW4=
kind: Secret
metadata:
  creationTimestamp: "2026-03-17T09:05:20Z"
  name: my-secret
  namespace: default
  resourceVersion: "7195"
  uid: 4040e4c7-a352-4931-b5db-9cf6238790e1
type: Opaque
```

Step 5: Use Secrets in Pods

Create a pod that uses the secret:

Create a file named **pod-with-secret.yaml** with the following content:

```
apiVersion: v1
kind: Pod
metadata:
  name: pod-with-secret
spec:
  containers:
  - name: mycontainer
    image: nginx
    env:
    - name: USERNAME
      valueFrom:
        secretKeyRef:
          name: my-secret
          key: username
    - name: PASSWORD
      valueFrom:
        secretKeyRef:
          name: my-secret
          key: password
```

Apply the YAML file:

```
kubectl apply -f pod-with-secret.yaml
```

```
type: Opaque
PS D:\kubs-misha> kubectl apply -f pod-with-secret.yaml
pod/pod-with-secret created
PS D:\kubs-misha> kubectl exec pod-with-secret -- printenv | grep -E 'USERNAME|PASSWORD
```

Verify the pod is running and the secret is being used:

```
kubectl exec pod-with-secret -- printenv | grep -E 'USERNAME|PASSWORD'
```

```
PS D:\kubs-misha> kubectl exec pod-with-secret -- printenv | Select-String "USERNAME|PASSWORD"

USERNAME=admin
PASSWORD=P@ssw0rd
```

Step 7: Clean Up

Delete the secrets:

```
kubectl delete secret my-secret

kubectl delete secret my-file-secret

kubectl delete secret my-registry-secret

kubectl delete secret my-manifest-secret
```

Delete the pod:

```
kubectl delete pod pod-with-secret
```

```
PS D:\kubs-misha> kubectl delete secret my-secret
secret "my-secret" deleted from default namespace
PS D:\kubs-misha> kubectl delete secret my-file-secret
secret "my-file-secret" deleted from default namespace
PS D:\kubs-misha> kubectl delete secret my-registry-secret
secret "my-registry-secret" deleted from default namespace
PS D:\kubs-misha> kubectl delete secret my-secret
Error from server (NotFound): secrets "my-secret" not found
PS D:\kubs-misha> kubectl delete pod pod-with-secret
pod "pod-with-secret" deleted from default namespace
PS D:\kubs-misha>
```